UNIT 4: STATISTICAL DATA

Simple Notes, Examples and Practice Questions

Based on the uploaded Unit 4 – Statistical Data PDF

This study material covers No-Code AI for Statistical Data, statistical concepts, MS Excel for statistical analysis, Orange Data Mining, AI Project Cycle mapping, and the Palmer Penguins case study.

1. No-Code AI for Statistical Data

What is No-Code AI?

No-Code AI is an approach that allows people to build or use AI and machine-learning models without writing programming code. It commonly uses visual interfaces, widgets, and drag-and-drop features.

Why is it useful?

- It reduces the need for programming knowledge.
- It makes AI more accessible to non-technical users.
- Visual and drag-and-drop features make the process easier to understand.
- Models can be created and tested faster.
- It can reduce development cost and the need for specialist coding staff.

Example: A zoo employee who does not know programming can use a No-Code AI tool to build a model that predicts future food prices for animals.

2. High-Code, Low-Code and No-Code

Approach	Main idea	Coding needed	Example/feature
High Code	Traditional software development.	A lot of manual coding.	Python, Java, C#.
Low Code	Uses visual interfaces and pre-built components, but some coding may be required.	Some coding.	Visual development with customization using code.
No Code	Uses visual interfaces and pre-built elements to create applications/models.	No coding required.	Drag-and-drop AI tools.

Simple way to remember: High Code = write most of the code; Low Code = write some code; No Code = use visual tools without coding.

3. Need for No-Code AI

- Coding can produce errors and debugging can be troublesome.
- No-Code tools avoid the need to write code for many tasks.
- They can save time and development cost.
- They allow non-technical professionals to use AI.

- They provide real-time visual feedback.

Who can use No-Code AI? Doctors, architects, musicians, educators, business users, students and other non-technical users can use such tools for suitable problems.

4. Advantages and Limitations of No-Code Tools

Advantages

- Accessibility – people with limited technical knowledge can use AI tools.
- Speed – models and workflows can often be built quickly.
- Ease of use – visual interfaces and drag-and-drop features simplify the process.
- Innovation – users can experiment with AI solutions for their own problems.

Limitations / disadvantages

- Lack of flexibility – users are limited by the features provided by the platform.
- Limited customization – highly specialized requirements may need coding.
- Automation bias – users may accept an automated recommendation even when contradictory information is correct.
- Security concerns – some No-Code platforms provide limited control over security and sensitive data.

5. Popular No-Code AI Tools

Azure Machine Learning: Cloud-based service provided by Microsoft; supports data preparation, model building, evaluation and deployment with visual tools.

Google Cloud AutoML: Allows users with limited ML knowledge to build custom machine-learning models with relatively little effort.

Orange Data Mining: Open-source data visualization, machine learning and data-mining toolkit using visual workflows.

Lobe: Visual machine-learning platform that helps users train custom models without traditional coding.

Teachable Machine: Web-based tool for creating models that can recognize images, sounds and poses.

6. Statistical Data and Important Concepts

Statistical Sampling

The entire set of raw data available for a test or experiment is called the **population**. A **sample** is a portion of that population selected for analysis.

Example: If a school has 1,000 students and we study the marks of 100 selected students, the 1,000 students are the population and the 100 students are the sample.

Descriptive Statistics

Descriptive statistics help us describe data and understand its main characteristics.

- **Mean:** The average of the values. Mean = Sum of all values ÷ Number of values.
- **Median:** The middle value after arranging the data from low to high.
- **Mode:** The value that occurs most often.

Example: For 1, 4, 5, 6, 4: Mean = 4, Median = 4, Mode = 4.

Distributions

A distribution shows how frequently values occur in a dataset. A normal distribution is approximately symmetrical, with most values concentrated around the central region.

Distributions can also be skewed when some values are much larger or smaller than most other values.

Probability

Probability is the likelihood that an event will occur. An event is an outcome of an experiment. Events may be independent or dependent.

Example: When a fair coin is tossed, the probability of getting heads is 1/2.

Variance and Standard Deviation

Variance measures how far values in a dataset are from the mean and is used to describe the spread of data. Standard deviation is a value that represents how widely the data values are distributed.

Outlier: A data point that lies at an abnormal distance from other values.

Example: In the dataset 10, 11, 12, 13, 100, the value 100 may be considered an outlier because it is far from the other values.

7. MS Excel for Statistical Data

The PDF demonstrates simple linear regression using speed and distance data in MS Excel.

Basic workflow:

- Enable the Analysis ToolPak: File → Options → Add-ins → Excel Add-ins → Go → select Analysis ToolPak.
- Identify the independent and dependent variables. In the example, Speed is X and Distance is Y.
- Select Speed and Distance and insert a Scatter Plot.
- Add a Linear Trendline to the scatter plot.
- Display the regression equation and R-squared value on the chart.
- Use Data → Data Analysis → Regression to obtain regression statistics.

- Use the generated linear equation, y = mx + c, to estimate distance for a given speed.

Example: If the regression equation is y = 5x + 2, then for x = 6, y = 5(6) + 2 = 32.

8. Orange Data Mining (ODM)

Orange Data Mining is an open-source data-mining and machine-learning software suite. It provides a graphical interface in which users build workflows using widgets and connect them on a canvas.

Key idea: Orange enables data analysis, visualization, preprocessing, modelling and evaluation using simple visual workflows without requiring traditional coding.

Widget group	Examples	Purpose
Data Loading	File, URL, Data Table	Load and view datasets.
Data Exploration	Scatter Plot, Distributions	Find patterns, relationships and distributions.
Preprocessing	Impute, Normalize, Select Columns	Handle missing values, scale data and choose columns.
Feature Selection	Select Columns, Select Best Features	Choose useful features for analysis.
Modelling	Classification Tree, k-Means, SVM, Logistic Regression	Build machine-learning models.
Evaluation	Test & Score, Cross Validation, ROC Curve	Measure model performance.
Visualization	Bar Chart, Heat Map, Scatter Plot	Present data visually.

9. AI Project Cycle in Orange Data Mining

The PDF maps the AI Project Cycle to steps carried out in Orange Data Mining.

Problem Scoping: Identify the problem and decide what the AI solution should predict or classify.

Data Acquisition: Collect or upload the required dataset. In Orange, the File widget can be used to load data.

Data Exploration: Clean missing data, select relevant columns and sample data when required.

Modelling: Choose and train an appropriate machine-learning model.

Evaluation: Use tools such as Test & Score to evaluate model performance.

Prediction / Deployment: Use the trained model to make predictions and apply the result to the problem.

10. Simple Orange Workflow Example

- Upload the dataset using the File widget.
- Use Data Table to inspect the dataset.
- Use preprocessing widgets if data needs cleaning.
- Select relevant columns/features.
- Choose a suitable model such as Linear Regression, Logistic Regression, Classification Tree or k-Means depending on the task.
- Connect the model to Test & Score to evaluate performance.
- Use Prediction to view predicted results.

Example – Food Price Prediction: The PDF uses a food price dataset and demonstrates selecting Food Price Index as the target variable, viewing the data, applying Linear Regression, evaluating the model and obtaining predictions. The result can help the zoo plan future animal-feed costs and fundraising.

11. Palmer Penguins Case Study

The Palmer Penguins case study demonstrates how Orange Data Mining can be used to predict the species of penguins from collected statistical features.

Problem: Researchers want to predict the species of Palmer Penguins based on collected data.

Useful features shown in the PDF include:

- Island
- Culmen length and culmen depth
- Flipper length
- Body mass
- Species as the target label

How AI can help: A machine-learning model can learn patterns in the measured features and use those patterns to predict the species of a new penguin.

Example: If a new penguin has particular flipper length, body mass and beak measurements, a trained classification model can use those features to predict its species.

12. Quick Revision Points

- No-Code AI allows AI workflows to be created without traditional programming.
- High Code requires extensive manual coding; Low Code requires some coding; No Code requires no coding.
- No-Code tools are accessible, fast and easy to use, but they may have limited flexibility and security control.
- Orange Data Mining is an open-source tool that uses widgets and drag-and-drop workflows.
- Population means the entire set of data under study; a sample is a portion of the population.
- Mean, median and mode are basic descriptive statistics.
- A distribution shows the frequency of values in a dataset.
- Probability describes the likelihood of an event occurring.
- Variance and standard deviation describe the spread of data.
- An outlier is a value unusually far from other observations.
- Scatter plots help visualize relationships between two variables.
- Linear regression can be represented by $y = mx + c$.
- Test & Score is used to evaluate model performance in Orange.
- Palmer Penguins is a classification case study using physical features to predict species.

13. Multiple Choice Questions (10)

1. Which approach allows users to build AI models without writing code?

- A. High Code
- B. Low Code
- C. No Code
- D. Machine Code

2. Which of the following is an example of a No-Code data-mining tool?

- A. Orange Data Mining
- B. C compiler
- C. Assembly language
- D. Text editor

3. Which measure represents the average of a set of values?

- A. Mode
- B. Mean
- C. Median
- D. Range

4. Which value occurs most often in a dataset?

- A. Mean
- B. Median
- C. Mode
- D. Variance

5. Which Orange widget is mainly used to display loaded data in tabular form?

- A. Data Table
- B. Test & Score
- C. k-Means
- D. ROC Curve

6. Which widget is used to evaluate the performance of a predictive model in Orange?

- A. File
- B. Data Table
- C. Test & Score
- D. Scatter Plot

7. Which graph is useful for showing the relationship between two numerical variables?

- A. Scatter Plot
- B. Pie chart
- C. Text box
- D. Data Table

8. Which of the following is a data point unusually far from other values?

- A. Mean
- B. Outlier
- C. Median
- D. Sample

9. In the Palmer Penguins case study, what is the target to be predicted?

- A. Body mass
- B. Island
- C. Penguin species
- D. Flipper length

10. Which of the following can be used for regression modelling in Orange?

- A. Linear Regression
- B. Data Table
- C. File
- D. Bar Chart

14. Subjective Questions (10)

1. Define No-Code AI. Explain why it is useful for statistical data analysis.

2. Differentiate between High Code, Low Code and No Code.

3. List any four advantages of using No-Code AI tools.

4. Explain any three limitations or challenges of No-Code AI tools.

5. What is statistical sampling? Differentiate between population and sample with an example.

6. Explain mean, median and mode with a suitable example.

7. What are variance, standard deviation and outliers? Explain their purpose in understanding data.

8. What is Orange Data Mining? Explain how its widget-based interface helps users analyse data.

9. Describe the steps for creating and evaluating a simple prediction model in Orange Data Mining.

10. Explain the Palmer Penguins case study and identify the features that can be used to predict penguin species.

15. Answer Key – MCQs

Question	Answer
1	C
2	A
3	B
4	C
5	A
6	C

7	A
8	B
9	C
10	A

Note: The notes and questions are based on the content of the uploaded Unit 4 PDF, including its explanations, examples, Orange Data Mining workflow, MS Excel activity, and Palmer Penguins case study.